



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI MAHADHEV FUEL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

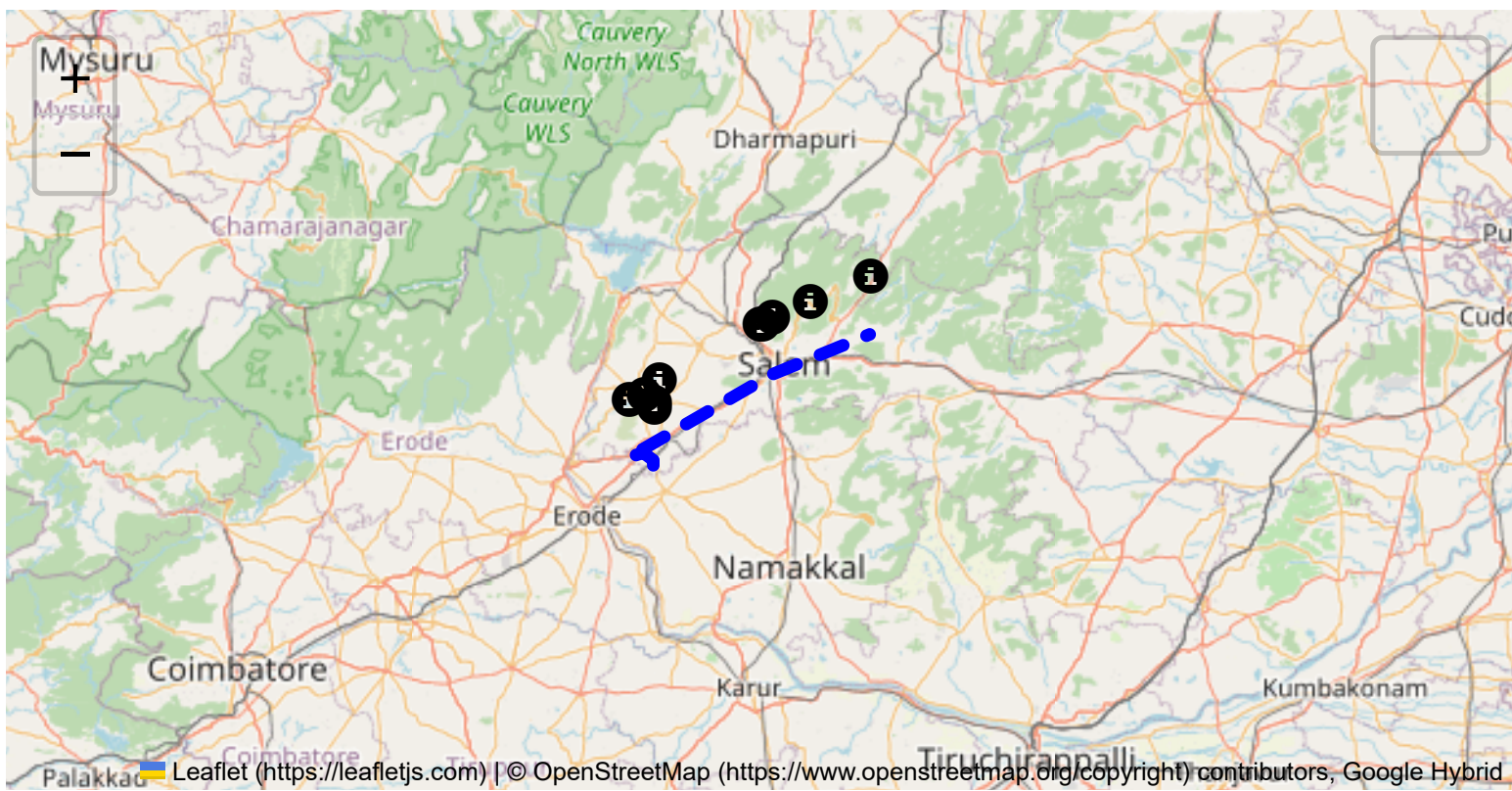
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

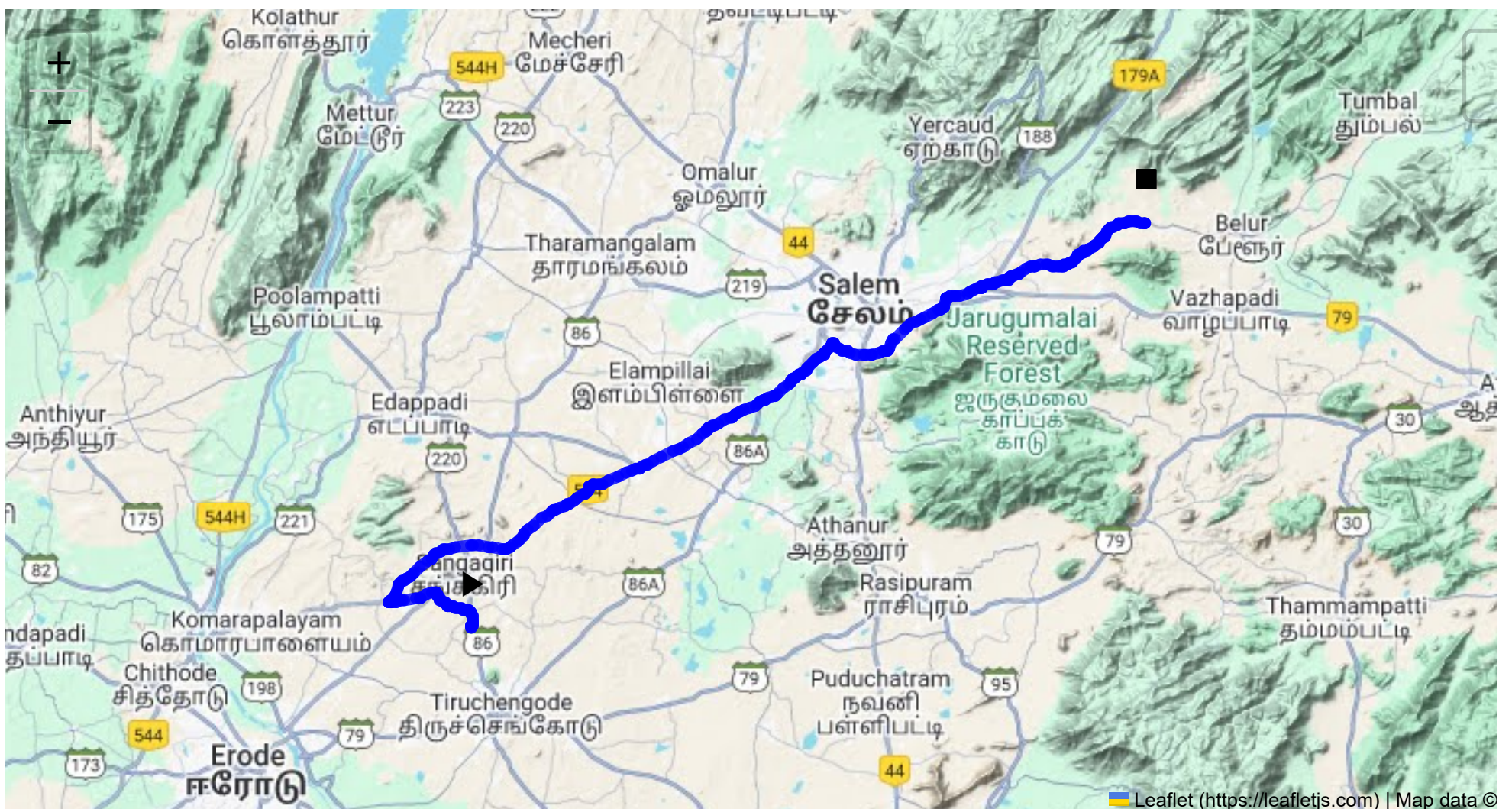
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 76.98 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours
Start: (11.4381, 77.8734)
End: (11.71581, 78.344906)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Salem covers approximately 76.98 kilometers. It begins close to Sangagiri and proceeds through various locations primarily along established roadways including the Sankari RS - Erode Privu Road, Bhavani Main Road, and NH544 before reaching the urban area of Salem. Key waypoints

include Sankari RS, Sankari West, and Padaiveedu, proceeding through Salem's Puthur Agraharam, Uthamasolapuram, Kondalampatti, and Udayapatti areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The typical weather conditions in these areas are generally hot and humid, particularly during the summer months (March to June). The monsoon season (July to September) may bring heavy rainfall, leading to potentially hazardous driving conditions such as waterlogged roads and reduced visibility. During the rainy season, caution is advised due to the risk of landslides and slippery roads.

3. Analysis of Traffic Patterns

Traffic patterns along this route suggest high congestion during peak hours, especially in areas near Salem. Morning (8 AM to 10 AM) and evening (5 PM to 7 PM) are particularly congested due to local commute and business hours. NH544, being a major highway, sees heavy vehicle traffic, which can lead to slow-moving traffic and occasional bottlenecks.

4. Assessment of Road Quality and Infrastructure

The road quality on NH544 is generally good, with well-maintained surfaces and clear signage. However, some local roads, particularly in rural areas near Padaiveedu and earlier in the journey, may experience potholes and narrower lanes. Regular maintenance is observed, but caution is advised due to occasional roadworks.

5. Suggestions for Alternative Routes for Emergencies

In case of road blockages or emergencies, alternative routes include taking State Highway 86, which runs parallel to NH544. It is advisable to remain updated on real-time traffic and road conditions through navigation apps to anticipate any reroutes.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to strict regulations in Tamil Nadu. Ensure proper labeling, documentation, and adherence to safety protocols. Night travel restrictions may apply for vehicles carrying certain hazardous materials, necessitating prior approvals and permits.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historically, there have been incidents involving overturned trucks and spills, particularly on highways due to overspeeding or poor load distribution. NH544's sharp turns and elevations necessitate cautious speed management, especially for heavy vehicles.

8. Environmental Considerations and Sensitive Areas

There are environmentally sensitive areas along the route, including agricultural lands. Emissions from trucks and the risk of hazardous material spills can impact these environments. Drivers should follow all guidelines for emission control and hazardous material handling to avoid contamination.

9. Analysis of Communication Coverage

Generally, communication coverage is strong along NH544 and in urban areas like Salem. However, segments along rural paths and minor roads may experience occasional dead zones. It's advisable to have a secondary communication device or pre-determined check-in points.

10. Estimated Emergency Response Times for Different Route Segments

In urban locations such as Salem, emergency response times are typically 15-30 minutes, while rural areas may see 30-60 minute response times due to distance and accessibility. NH544 has quicker response times due to its frequent patrols and proximity to emergency services.

11. Overall Summary of Risk Assessment

The overall risk assessment categorizes this route as moderate for transporting hazardous materials. Factors including traffic congestion, varying road conditions, and weather hazards increase the complexity of travel. Proper planning, real-time monitoring, and adherence to safety regulations can mitigate risks significantly. Ensuring vehicle maintenance and driver readiness is crucial for traversing efficiently and safely.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44900, 77.87406	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	High	11.45352, 77.85698	15 KM/Hr
7	Turn	Medium	11.45847, 77.85140	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46314, 77.84928	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	High	11.45634, 77.81673	15 KM/Hr
12	Turn	High	11.61233, 78.10397	15 KM/Hr
13	Turn	High	11.61216, 78.10423	15 KM/Hr
14	Turn	High	11.63300, 78.12746	15 KM/Hr
15	Turn	Medium	11.65627, 78.20215	30 KM/Hr
16	Turn	Medium	11.65664, 78.20240	30 KM/Hr
17	Turn	Medium	11.65825, 78.20469	30 KM/Hr
18	Turn	High	11.66380, 78.20648	15 KM/Hr
19	Turn	Medium	11.67128, 78.22971	30 KM/Hr
20	Turn	High	11.67111, 78.23134	15 KM/Hr
21	Turn	Medium	11.67195, 78.23146	30 KM/Hr
22	Turn	Medium	11.67666, 78.24906	30 KM/Hr
23	Turn	Medium	11.68494, 78.26198	30 KM/Hr
24	Turn	Medium	11.68551, 78.26771	30 KM/Hr
25	Turn	Medium	11.68751, 78.27618	30 KM/Hr
26	Turn	Medium	11.68624, 78.28833	30 KM/Hr
27	Turn	Medium	11.68848, 78.29399	30 KM/Hr
28	Turn	Medium	11.70555, 78.31562	30 KM/Hr
29	Turn	Medium	11.70752, 78.31775	30 KM/Hr
30	Blind Spot	Blind Spot	11.71593, 78.34499	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
19	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
20	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
21	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
22	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
23	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
24	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
25	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
28	school	Govt Hr Sec School	11.6754733, 78.2459724	30 km/h	Medium
29	school	St Mary's Hr Sec School	11.6872096, 78.2796818	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

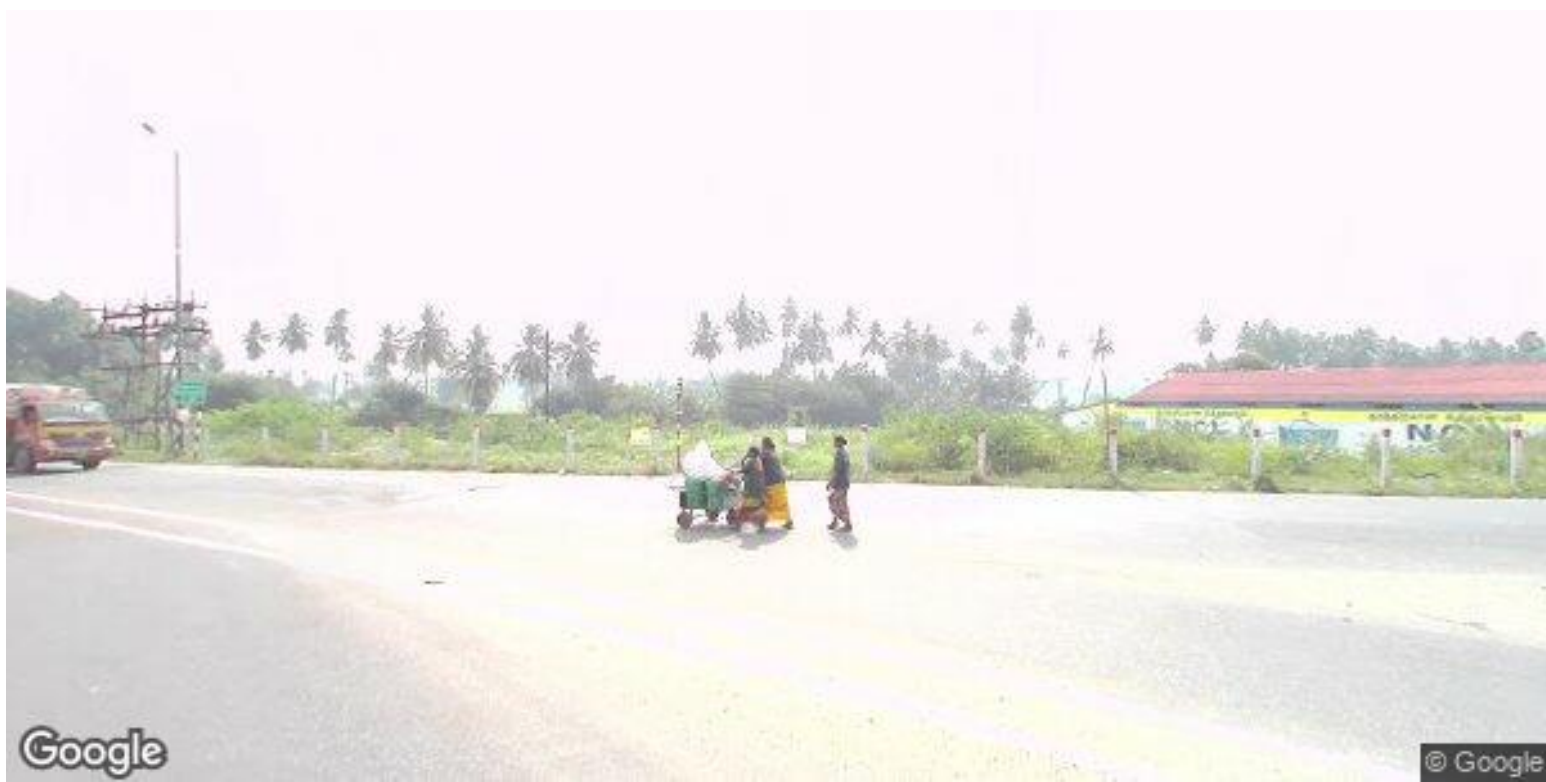


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61233, 78.10397



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.61216, 78.10423



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63300, 78.12746



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65627, 78.20215



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65664, 78.20240



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65825, 78.20469



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66380, 78.20648



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67128, 78.22971



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67111, 78.23134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67195, 78.23146



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67666, 78.24906



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68494, 78.26198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68551, 78.26771



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68751, 78.27618



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68624, 78.28833



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68848, 78.29399



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70555, 78.31562



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70752, 78.31775
